

Jialeng Ni

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EDUCATION

University of Michigan — M.S. in Robotics

Ann Arbor, MI | Expected Graduation: December 2026

University of Michigan — B.S. in Computer Science and Engineering

Ann Arbor, MI | Aug 2023 – May 2025

GPA: 3.90/4.00 | Summa Cum Laude | Jackson and Muriel Lum Undergraduate Scholarship

Shanghai Jiao Tong University — B.S. in Electrical and Computer Engineering

Shanghai, China | Sep 2021 – Aug 2025

GPA: 3.81/4.00 | Distinguished Graduate | Undergraduate Outstanding Scholarship

WORK EXPERIENCE

XPENG Robotics (U.S.)

May 2026 – Aug 2026

Machine Learning Engineer Intern

Santa Clara, CA

IronMind: World Model for Embodied Robotic Manipulation

World Model Team | Jul 2026 – Aug 2026

- Built a MoT world model for robotic manipulation with large-scale egocentric pretraining in the camera frame; analyzed how camera- versus torso-frame representations affect pretraining and downstream post-training. Ablated DINO/JEPA features, VAE latents, and depth maps, and extended monocular inputs to stereo using cross-view disparity for implicit depth.
- Optimized low-latency inference and action execution to mitigate observation-action misalignment and closed-loop lag, introducing dynamic latency compensation, temporal aggregation, and action-chunk scheduling.

Real-World Online Reinforcement Learning for Humanoid Robots

General Manipulation Team | May 2026 – Jul 2026

- Built an end-to-end real-world online RL pipeline for an in-house Vision-Language-Action (VLA) model using Diffusion Steering and off-policy Soft Actor-Critic (SAC), with asynchronous rollouts, experience replay, policy updates, evaluation, deployment, and a high-throughput replay buffer.
- Built a real-robot safety stack with exploration constraints, safety shielding, abnormal-termination handling, state recovery, and checkpoint-based resumption for long-duration training.
- Proposed Projected Steerable Subspace (PSS), extracting action-sensitive directions from state-conditioned diffusion-noise perturbations and compressing the SAC actor's full-dimensional noise control into a low-dimensional subspace; improved training efficiency by 2–5x.

Lightwheel Inc.

Oct 2025 – Feb 2026

Applied Research Intern

Remote

Large Reward Model for General-Purpose Robotic Manipulation

- Built a Rewind-based data pipeline over 100K+ human-teleoperation episodes, using custom sampling to generate positive and negative examples for manipulation value estimation.
- Fine-tuned Qwen3-VL-2B-Instruct with LoRA for trajectory-level reward/value estimation from task descriptions and one-shot successful references; used Advantage-Weighted Regression (AWR) to optimize a 3B-parameter Pi0 VLA policy, improving task success by about 30% over behavior cloning and enabling failure detection and large-scale data quality control.

Shanghai AI Laboratory — OpenRobotLab

May 2025 – Aug 2025

Research Intern

Shanghai, China

InternManip: Unified Robot Manipulation Learning Framework

GitHub ↗ | Challenge ↗

- Co-developed a unified training and evaluation framework supporting ACT, Diffusion Policy, Pi0, and GR00T-N1/N1.5 across GenManip-v1, CALVIN, and SimplerEnv.
- Co-organized the IROS 2025 Multimodal Robot Learning Challenge across InternUtopia simulation and real-world tracks.

RESEARCH EXPERIENCE

NeuralMetric: Transformer-Based Safety Metric for Autonomous Driving

Feb 2024 – Sep 2024

McCity, University of Michigan

Ann Arbor, MI

- Developed a Transformer-based safety metric that learned context-dependent collision risk from trajectories; built three simulation datasets with realistic accident distributions and mitigated severe class imbalance using weighted losses and sampling.
- Achieved state-of-the-art predictive accuracy across highways, intersections, and roundabouts with 2 ms GPU / 76 ms CPU latency, delivering a 100–1000x speedup over prior methods.

Publication: “NeuralMetric: An Accurate and Efficient Real-Time Safety Metric for Automated Driving Systems,” *Transportation Research Part A: Policy and Practice*, 2026.

PROJECTS & SKILLS

Independent Open-Source Project

PiperPilot

github.com/tomakeIT/PiperPilot ↗ | jialengni.com/PiperPilot ↗

- Independently built an open-source, end-to-end robot learning stack for AgileX Piper/Piper-X, supporting Quest 3 and SpaceMouse teleoperation, LeRobot dataset collection/replay, policy training, evaluation, and real-robot deployment.

Programming & Tools: Python, PyTorch

Robot Learning: RL, Imitation Learning, VLA Models, World Models, Reward Modeling